Decision and Risk Analysis

Practice Questions for Exam 2

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- Conceptual questions
- Open questions (for more understanding)
- Calculations (really short hand calculations)
- **Answers are at the end.**

Chapter 6

Approximating the Posterior

Question 6.1

For each MCMC diagnostic, say in plain words **what question it answers** and **what a good result looks like**.

1. Trace plot
2. Density plots of the parallel chains, overlaid
3. Effective sample size ratio N_{eff}/N
4. $\hat{R}$

Question 6.2

A grid of 101 values approximates the Beta(11, 3) posterior almost perfectly, while a 6-point grid only produced draws at 0.4, 0.6 and 0.8.

(a) Why was the 6-point grid so poor?

(b) If we keep 101 grid values *per parameter*, how many posterior evaluations would a 3-parameter model need? A 10-parameter model?

(c) What does this suggest about when to use MCMC instead?

Question 6.3

Unlike grid approximation, MCMC draws are *dependent* (each depends on the previous one) and are *not* drawn directly from the posterior.

Why can we still use a Markov chain sample to estimate the posterior mean or a credible interval? What role does running the chain longer (and discarding burn-in) play?

Question 6.4

You run 4 parallel chains. Each trace plot, viewed *alone*, looks like stable white noise. But chains 1–2 hover around $\theta \approx 0.3$ and chains 3–4 hover around $\theta \approx 0.7$.

Would $\hat{R}$ be close to 1? Why do single-chain checks miss this problem, and what should you do?

Question 6.5

To "reduce autocorrelation," a colleague keeps every 10th draw, cutting a fast-mixing `rstan` sample from 20,000 to 2,000 values.

Is this a good idea? What is being traded off, and when (if ever) is thinning worth it?

Question 6.6

$Y \mid \pi \sim \text{Bin}(2, \pi)$, we observe $Y = 2$, and the prior is Beta(2, 2), so $f(\pi) \propto \pi(1 - \pi)$. Use the grid $\pi \in \{0.25,\ 0.50,\ 0.75\}$.

(a) At each grid value compute prior $\times$ likelihood.

(b) Normalize to get the grid posterior probabilities.

(c) Compute the grid posterior mean and compare with the exact posterior Beta(4, 2), mean 4/6. Why do they differ?

You may drop constant factors (the 6 in the Beta(2, 2) density) that are the same for every π, so use $\pi(1 - \pi) \cdot \pi^2$.

$$\text{Prior: } f(\pi) = 6\,\pi^{2-1}(1-\pi)^{2-1} \propto \pi(1-\pi)$$

$$\text{Likelihood: } L(\pi \mid y=2) = \binom{2}{2}\pi^2(1-\pi)^0 = \pi^2$$

$$\text{Posterior: } f(\pi \mid y) \propto \pi(1-\pi)\cdot\pi^2 = \pi^3(1-\pi) = \pi^{4-1}(1-\pi)^{2-1}$$

$$\Rightarrow\ \pi \mid y \sim \text{Beta}(4,2)$$

Question 6.7

You run 4 chains with 5,000 draws each (after burn-in). The variance of all draws combined is 0.09, and the typical variance within a single chain is 0.04. Compute $\hat{R} \approx \sqrt{\text{Var}_{\text{combined}}/\text{Var}_{\text{within}}}$ and interpret it.

Chapter 7

MCMC under the Hood

Question 7.1

(a) Name the two steps of every Metropolis-Hastings iteration.

(b) If the proposal is rejected, what is the chain's next value? Does that value count as a draw?

(c) Why do these repeated values matter for getting the posterior right?

Question 7.2

In the Normal-Normal example (posterior $\mu \mid y \sim N(4, 0.6^2)$), the Metropolis algorithm sometimes *accepts* a proposal that is less plausible than the current value.

Why is that necessary? Describe the sample you would get from a rule that only accepts moves to more plausible values.

Question 7.3

The acceptance probability uses the *unnormalized* posterior $f(\mu)L(\mu \mid y)$, not $f(\mu \mid y)$.

Why is this enough? And why is this the key reason MCMC works for models where grid approximation and exact formulas fail?

Question 7.4

Uniform proposal $\mu' \sim \text{Unif}(\mu - w,\ \mu + w)$. Compare $w = 0.01$, $w = 1$ and $w = 100$.

(a) Which has the highest acceptance rate?

(b) Sketch / describe the trace plot for $w = 0.01$ and $w = 100$.

(c) Why is a very high acceptance rate *not* a sign of a good chain?

Question 7.5

For $\pi \in (0, 1)$ we use an independence sampler: every proposal comes from the same Beta(a, b) model, regardless of the current π. The acceptance probability is

$$\alpha = \min\left\{1,\ \frac{f(\pi' \mid y)}{f(\pi \mid y)} \cdot \frac{q(\pi)}{q(\pi')}\right\}.$$

(a) Why is the correction $q(\pi)/q(\pi')$ needed if we propose from, say, Beta(5, 1)?

(b) What happens to the correction with a Beta(1, 1) proposal?

Question 7.6

The posterior is $\mu \mid y \sim N(0,\ 1)$: most plausible at $\mu = 0$. Metropolis proposes $\mu' \sim \text{Unif}(\mu - 3,\ \mu + 3)$ (symmetric), then accepts it with probability α.

(a) The chain is at $\mu = 0$ (the peak) and proposes $\mu' = 2$. Is 2 more or less plausible than 0? Compute α.

(b) The chain is at $\mu = 2$ and proposes $\mu' = 0$. Compute α *without* a calculator. Why?

(c) Out of 100 times the chain proposes these moves, about how many times will it actually move $0 \to 2$? And $2 \to 0$? What does that say about how often the chain visits 2 relative to 0?

Formulas on the next slide.

Formulas (for Question 7.6)

$$\text{Posterior (kernel):}\quad f(\mu \mid y) \propto \exp\left(-\frac{\mu^2}{2}\right)$$

$$\text{Ratio:}\quad \frac{f(\mu' \mid y)}{f(\mu \mid y)} = \exp\left(-\frac{\mu'^2 - \mu^2}{2}\right)$$

$$\text{Acceptance (symmetric):}\quad \alpha = \min\left\{1,\ \frac{f(\mu' \mid y)}{f(\mu \mid y)}\right\}$$

$$\text{Useful value:}\quad e^{-2} \approx 0.135$$

Rule of thumb: moving to a *more* plausible value is always accepted ($\alpha = 1$); moving to a *less* plausible value is accepted with probability equal to the plausibility ratio.

Question 7.7

Posterior: $\mu \mid y \sim N(4, 0.6^2)$, so $f(\mu \mid y) \propto \exp\left(-\frac{(\mu-4)^2}{2(0.36)}\right)$.

Symmetric Uniform proposals.

The chain is at $\mu = 3.7$ and proposes $\mu' = 4.9$.

(a) Compute the acceptance probability α (hint: $e^{-1} \approx 0.368$).

(b) To decide, we draw $U \sim \text{Unif}(0, 1)$ and accept if $U < \alpha$. What is the next value if $U = 0.50$? If $U = 0.20$?

(c) What would α be for the reverse move, $4.9 \to 3.7$?

Chapter 8

Posterior Inference & Prediction

Question 8.1

MoMA example: π = proportion of artists who are Gen X or younger. Label each question as **estimation**, **hypothesis testing**, or **prediction**:

1. What is a plausible range for π?
2. Is π below 0.20?
3. How many of the next 20 artworks will be by Gen X artists?
4. Is the typical value of π closer to 0.15 or 0.30?

Question 8.2

The middle 95% credible interval for π is $(0.10,\ 0.24)$.

(a) Will the middle 99% interval be wider or narrower? The middle 50%?

(b) Which credible level is the "correct" one to report?

Question 8.3

MoMA example: prior Beta(4, 6), $Y = 14$ of $n = 100$ artists are Gen X or younger $\Rightarrow$ posterior Beta(18, 92). The middle 95% credible interval for π is $(0.10,\ 0.24)$.

A student says: *"If we repeated the study many times, 95% of such intervals would contain π."*

Is this the Bayesian interpretation? State it correctly. When would a highest posterior density interval differ from this one?

Question 8.4

Testing $H_a : \pi < 0.2$:

	Prior	Posterior
$P(H_a)$	0.086	0.849
Odds	0.094	5.62

Bayes factor ≈ 60.

What does each number measure? Why can the Bayes factor be huge while the posterior probability is "only" 0.85?

Question 8.5

We want to test $H_0 : \pi = 0.3$ vs. $H_a : \pi \neq 0.3$.

(a) Why can't we compute $P(H_0 \mid y)$ or a Bayes factor directly?

(b) What is the practical workaround, and what does it conclude here?

(c) Why should that workaround be set *before* seeing the data?

Question 8.6

Predict $Y' =$ number of Gen X artists among 20 new works. A student uses the posterior mean $\hat{\pi} = 0.164$ and reports $Y' \sim \text{Bin}(20,\ 0.164)$.

(a) Which source of variability is missing?

(b) Will the student's predictions be too narrow or too wide?

(c) With MCMC, why do we simulate one $Y_i' \sim \text{Bin}(20, \pi_i)$ per draw instead of using $20\,\pi_i$?

Question 8.7

For $H_0 : \pi \geq 0.2$ vs. $H_a : \pi < 0.2$:

- Frequentist p-value $= P(Y \leq 14 \mid \pi = 0.2) \approx 0.08$
- Bayesian $P(H_a \mid Y = 14) \approx 0.849$

A student concludes: *"There is an 8% chance that H_0 is true."*

What is wrong? What is the probability of H_0 according to the Bayesian analysis, and what does each method condition on?

Question 8.8

Prior $\pi \sim \text{Beta}(4, 6)$; data $Y = 14$ of $n = 100$.

Recall: $\text{Beta}(\alpha, \beta)$ has mean $\dfrac{\alpha}{\alpha + \beta}$ and mode $\dfrac{\alpha - 1}{\alpha + \beta - 2}$.

(a) What are the posterior parameters?

(b) Compute the posterior mean and mode.

(c) Compute $E(Y' \mid Y = 14)$, the expected number of Gen X artists among 20 new works. Why can you get it without the full posterior predictive model?

Question 8.9

Recall: $\text{odds} = \dfrac{p}{1-p}$, $\quad \text{BF} = \dfrac{\text{posterior odds}}{\text{prior odds}}$.

(a) A hypothesis has prior probability 0.2 and posterior probability 0.8. Compute the prior odds, posterior odds, and Bayes factor.

(b) MoMA: prior odds of $H_a : \pi < 0.2$ are 0.094 and the Bayes factor is 60. Compute the posterior odds and the posterior probability $P(H_a \mid Y = 14)$.

Chapter 9

Simple Normal Regression

Question 9.1

Bike model: posterior median slope $\beta_1 \approx 82.2$, 80% credible interval $(75.6,\ 88.8)$, and $P(\beta_1 > 0 \mid \text{data}) \approx 1$.

(a) Interpret β_1 in one sentence.

(b) Does this show that warmer weather *causes* more ridership? What else could be going on?

Question 9.2

Bike model: $Y_i \sim N(\beta_0 + \beta_1 X_i,\ \sigma^2)$, $X =$ temperature (°F). The posterior median line is

$$\hat{\mu} = -2194 + 82.2\, X.$$

(a) Should we worry that the model predicts -2194 rides?

(b) Why did we put the prior on the *centered* intercept β_{0c} (rides at an average temperature) instead of β_0?

Question 9.3

From the posterior we get two 95% intervals for 75°F:

- Credible interval for the *average* ridership on 75°F days: (3843, 4095)
- Prediction interval for ridership on *one* 75°F day: (1500, 6482)

(a) Why is the second so much wider?

(b) A manager must decide how many bikes to stock for tomorrow (75°F). Which interval is relevant?

(c) With 50,000 days of data, what would happen to each?

Question 9.4

Data arrive in phases: 30 days, 60 days, then all 500 days. After the first 30 days (all cold winter days), the posterior for β_1 is still vague and does not reveal the warm-weather pattern. By 500 days it is concentrated near 80.

Why are the first 30 days so uninformative about the slope, even though 30 is not a tiny sample? What does this show about how a posterior evolves?

Question 9.5

(a) Prior $\beta_1 \sim N(100, 40^2)$. Using mean $\pm$ 2 SD, what range of slopes does the prior consider plausible? What does that mean in rides?

(b) Prior $\sigma \sim \text{Exp}(0.0008)$. What is the prior mean of σ? (Exp(λ) has mean $1/\lambda$.)

(c) With the posterior median line $-2194 + 82.2X$: predict ridership at 75°F, and the expected difference between a 60°F and a 70°F day.

Question 9.6

Three posterior draws, each with a standard Normal value z:

Draw	β_0	β_1	σ	z
1	-2200	82	1300	0.5
2	-2100	81	1250	-1.0
3	-2300	84	1280	0.2

(a) For each draw compute $\mu = \beta_0 + 75\beta_1$ and $Y' = \mu + \sigma z$.

(b) Which set of values (μ's or Y''s) builds the credible interval for the *average* 75°F day, and which builds the prediction interval? Which set is more spread out, and why?

Chapter 15

Hierarchical Models

Question 15.1

Which data sets are grouped (hierarchical)? Name the group, and say why observations within a group are likely correlated.

1. Test scores of students from 12 schools
2. Blood pressure measured monthly on 40 patients
3. One survey response from each of 500 randomly sampled adults
4. Daily high temperatures in 10 cities for a year

Question 15.2

Cherry Blossom data: 252 race times from 36 runners (50s–60s), each observed in several years.

Complete pooling (one regression of time on age): slope ≈ 0.27 min/year, 80% CI $(-0.30,\ 0.84)$. Yet nearly *every* runner gets slower with age, faster than this.

How can pooling all the data hide a trend that almost every individual shows? Which assumption is violated?

Question 15.3

A no-pooled model fits each runner separately. Runner 1 has only 5 races and improved over them; the model predicts a time under 75 minutes at age 62. The book considers 75–85 minutes more reasonable.

(a) Why is the no-pooled prediction implausible?

(b) Why can't a no-pooled model predict anything for a new runner (e.g. *you*)?

Question 15.4

Runner 29 is fast and very consistent; runner 17 is slow and highly variable.

(a) Distinguish *within-group* and *between-group* variability using these runners.

(b) Why are a runner's own race times correlated?

(c) How does a hierarchical (partial pooling) model use both kinds of variability to sit between the other two approaches?

Question 15.5

Two runners, three races each:

	Ages	Times (min)
Runner A	55, 56, 57	80, 81, 82
Runner B	60, 61, 62	70, 71, 72

(a) No pooling: what is each runner's slope (min per year)?

(b) Complete pooling: compute the least-squares slope through all 6 points, $b = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$.

(c) Explain the contradiction.

Question 15.6

A useful simplification: a hierarchical estimate for a runner is roughly

$$w \times (\text{runner's own mean}) + (1 - w) \times (\text{overall mean}), \quad 0 \leq w \leq 1.$$

Overall mean: 90 min. Runners X and Y both have own mean 80 min; X ran 2 races, Y ran 10.

(a) Compute the estimate with $w = 0.5$ and with $w = 0.9$.

(b) Which w fits X and which fits Y? Why?

(c) Which values of w give complete pooling and no pooling?

Multiple Choice

MCQ 1

You run 4 parallel chains. Which diagnostic most directly detects that the chains are exploring *different* regions of the posterior?

(A) The effective sample size ratio N_{eff}/N

(B) The autocorrelation plot of chain 1

(C) $\hat{R}$

(D) The trace plot of a single chain

MCQ 2

Metropolis algorithm with a symmetric proposal. The unnormalized posterior is 0.4 at the current value and 0.1 at the proposal. What is the probability of accepting the proposal?

(A) 0.10

(B) 0.25

(C) 0.40

(D) 1 — it cannot be computed without normalizing

MCQ 3

Which statement about the 95% posterior prediction interval for ridership on a single 75°F day is correct?

(A) It is narrower than the 95% credible interval for the *average* ridership on 75°F days.

(B) It reflects only uncertainty in β_0 and β_1.

(C) It reflects both uncertainty in the parameters and day-to-day variability around the regression line.

(D) With enough data it shrinks to a single number.

MCQ 4

Using the Cherry Blossom data, we want to predict race times for a *new* runner who is not in the sample. Which models can do this?

(A) Only the no-pooled model

(B) Only the complete-pooled model

(C) The complete-pooled and hierarchical (partial pooling) models

(D) All three models

Answers

Answer 6.1

1. **Trace plot:** is the chain wandering stably? Good: looks like random noise around a stable level, with no trends or flat stretches.
2. **Overlaid densities:** do the chains agree? Good: the density curves nearly coincide.
3. $N_{\mathbf{eff}}/N$**:** how many independent draws are our dependent draws worth? Good: not too small (above ≈ 0.1).
4. $\hat{R}$**:** is the variability across chains similar to the variability within chains? Good: close to 1 (below ≈ 1.05).

Answer 6.2

(a) The posterior can only put mass on grid points. With 6 points, values between them (e.g. 0.7–0.9, where Beta(11, 3) is concentrated) can't be represented, so the approximation is crude.

(b) $101^3 \approx 1.03$ million evaluations; $101^{10} \approx 1.1 \times 10^{20}$. Impossible.

(c) The *curse of dimensionality*: grids waste effort on implausible regions. MCMC spends its time where the posterior has mass, so it scales to many parameters.

Answer 6.3

- The chain is built so that, in the long run, the *distribution of visited values* converges to the posterior.
- Averages over the chain (mean, quantiles) therefore approximate posterior quantities.
- Dependence doesn't make the estimates wrong; it makes each draw carry *less information* (see effective sample size).
- Longer chains: more information and time to settle. Burn-in: discard early values that still reflect the arbitrary starting point.

Answer 6.4

No! $\hat{R}$ would be far above 1 (well past 1.05).

- $\hat{R} \approx \sqrt{\text{Var}_{\text{combined}}/\text{Var}_{\text{within}}}$. The combined chains vary much more than any single chain.
- Each chain looks stable because it is stuck in its own region; only *comparing* chains reveals they disagree.
- Action: don't trust the results. Run longer, use dispersed starting values, and check the model (e.g. a multimodal or poorly identified posterior).

Answer 6.5

Usually not.

- Thinning lowers autocorrelation but *throws away* information: 20,000 correlated draws generally beat 2,000 less correlated ones.
- The chain is already fast-mixing, so there is little to gain.
- `rstan`/`rstanarm` (HMC) are efficient; their authors advise thinning only when memory/storage is a problem.
- To fix slow mixing, prefer running *longer chains* (or improving the model).

Answer 6.6

π	$\pi^3(1-\pi)$	$\times 256$	Posterior
0.25	0.0117	3	$3/46 \approx 0.065$
0.50	0.0625	16	$16/46 \approx 0.348$
0.75	0.1055	27	$27/46 \approx 0.587$
		46	1

(c) Mean $= (0.25 \cdot 3 + 0.50 \cdot 16 + 0.75 \cdot 27)/46 = 29/46 \approx 0.630$, vs. exact 0.667.

The grid can't put mass above 0.75, where Beta(4, 2) still has a lot of its mass (its mode is 0.75), so the coarse grid pulls the mean down. A finer grid fixes this.

Answer 6.7

$\hat{R} \approx \sqrt{0.09/0.04} = \sqrt{2.25} = 1.5$. Far above 1 (or even a upper limit of 1.05): the chains together vary much more than each one alone, so they disagree. The simulation is unstable and its results should not be trusted.

Answer 7.1

(a) (1) *Propose* a new value from a proposal model. (2) *Decide*: accept it with probability α, otherwise stay.

(b) The chain stays put: the next value equals the current value, and yes, it counts as a draw in the sample.

(c) Repeats are how the chain gives more weight to plausible values: it "lingers" where the posterior is high. Dropping the repeats would over-represent less plausible values and distort the approximation.

Answer 7.2

- The goal is not to *find* the most plausible value but to visit every value *in proportion to* its plausibility.
- Less plausible regions (the tails) must be visited sometimes, just less often. Accepting with probability equal to the plausibility ratio does exactly that.
- An "uphill only" rule acts like an optimizer: the chain climbs to $\mu \approx 4$ and stays there. The sample has almost no spread, so credible intervals would be far too narrow.

Answer 7.3

- α depends only on a *ratio*:

$$\frac{f(\mu' \mid y)}{f(\mu \mid y)} = \frac{f(\mu')L(\mu' \mid y)/f(y)}{f(\mu)L(\mu \mid y)/f(y)} = \frac{f(\mu')L(\mu' \mid y)}{f(\mu)L(\mu \mid y)}.$$

  The normalizing constant $f(y)$ cancels.
- $f(y)$ is the hard part: a (multiple) integral over all parameters. Prior $\times$ likelihood is easy to evaluate at any point.
- So MCMC needs only what we can always compute, even in models with many parameters.

Answer 7.4

(a) $w = 0.01$: proposals are nearly identical to the current value, so almost all are accepted.

(b) $w = 0.01$: a slowly wandering, smooth trace that explores only a small region (high autocorrelation). $w = 100$: long flat stretches where the chain is stuck, because most proposals land in implausible regions and are rejected.

(c) Accepting tiny steps isn't exploring. Efficiency depends on *how far* the chain moves, not just *how often*. $w = 1$ balances step size and acceptance (the "Goldilocks" choice).

Answer 7.5

(a) Beta(5, 1) proposes large π values much more often than small ones. Without correction, the chain would over-visit large values just because they are proposed more, biasing the sample toward the proposal. The factor $q(\pi)/q(\pi')$ penalizes moves to frequently proposed values and boosts moves to rarely proposed ones.

(b) Beta(1, 1) is Uniform: $q(\pi) = q(\pi') = 1$, so the correction equals 1 and the algorithm reduces to the plain Metropolis ratio.

Answer 7.6

(a) 2 is farther from the peak, so *less* plausible:

$$\frac{f(2 \mid y)}{f(0 \mid y)} = \exp\left(-\frac{2^2 - 0^2}{2}\right) = e^{-2} \approx 0.135 \;\Rightarrow\; \alpha \approx 0.135$$

(b) 0 is the peak, so it is *more* plausible than 2: the ratio is $e^2 > 1$, so $\alpha = 1$. No calculation needed: uphill moves are always accepted.

(c) About $100 \times 0.135 \approx 13$ moves $0 \to 2$, and all 100 moves $2 \to 0$.

The chain moves from 2 back to 0 far more readily than from 0 out to 2. So in the long run it spends about 0.135 times as much time at 2 as at 0, which is exactly $f(2 \mid y)/f(0 \mid y)$. The chain visits each value in proportion to its posterior plausibility.

Answer 7.7

(a) Symmetric proposal, so α is the posterior ratio:

$$\frac{f(4.9 \mid y)}{f(3.7 \mid y)} = \exp\left(-\frac{0.9^2 - 0.3^2}{2(0.36)}\right) = e^{-0.72/0.72} = e^{-1} \approx 0.368.$$

So $\alpha = \min\{1, 0.368\} = 0.368$.

(b) $U = 0.50 > 0.368$: reject, next value is 3.7.
$U = 0.20 < 0.368$: accept, next value is 4.9.

(c) 3.7 is more plausible than 4.9 (closer to 4), so the ratio is $e^1 > 1$ and $\alpha = 1$: always accepted.

Answer 8.1

1. Plausible range for π: **estimation** (credible interval).
2. Is $\pi < 0.20$? **Hypothesis testing** (posterior probability, odds, Bayes factor).
3. Next 20 artworks: **prediction** (posterior predictive model for Y').
4. Typical value of π: **estimation** (posterior mean, median or mode).

Note: 1, 2 and 4 are about the parameter π; 3 is about future *data*.

Answer 8.2

(a) 99%: **wider**, because it must hold more posterior probability: (0.085, 0.265).
50%: **narrower**: (0.139, 0.186).

(b) None. 95% is a convention, not a rule. The level should fit the purpose, and reporting several levels (or the whole posterior) communicates uncertainty better than any single interval.

Answer 8.3

Nope. That is the frequentist confidence-interval interpretation (about the procedure over repeated samples).

Bayesian: given the prior and the observed data, there is a 95% posterior probability that π is between 0.10 and 0.24. The probability statement is about π, not about the interval.

HPD vs. middle: they differ when the posterior is strongly *skewed*. HPD keeps the most plausible values; the middle interval cuts equal tail areas. Beta(18, 92) is fairly symmetric, so they are nearly the same here.

Answer 8.4

- **Posterior probability 0.849:** our *final* belief in H_a, combining prior and data.
- **Bayes factor** $= \dfrac{\text{posterior odds}}{\text{prior odds}} = \dfrac{5.62}{0.094} \approx 60$: how much the *data* shifted the odds. A measure of evidence, not of belief.
- The prior was skeptical of H_a (only 0.086). The data raised the odds 60-fold, but starting from a low base, the posterior ends at 0.85.
- Read both with nuance; avoid rigid "significant" cutoffs.

Answer 8.5

(a) π is continuous, so $P(\pi = 0.3) = 0$ under both prior and posterior. Posterior odds and Bayes factor are undefined (0 / 0).

(b) Define a practically meaningful buffer, e.g. $H_0 : \pi \in (0.25, 0.35)$. The whole buffer lies above the 95% CI $(0.10, 0.24) \Rightarrow$ strong evidence against H_0.

(c) Choosing the buffer after seeing the data lets us move the goalposts to reach a desired conclusion. It should reflect subject-matter meaning, not results.

Answer 8.6

(a) *Posterior variability*: uncertainty in π itself. Plugging in $\hat{\pi}$ keeps only sampling variability.

(b) Too narrow (overconfident). Same center (≈ 3.3), but

	$\mathrm{Var}(Y')$
Bin(20, 0.164)	2.74
Posterior predictive (Beta-Binomial)	3.21

(c) $20\,\pi_i$ is an expected count and drops sampling variability. Drawing $Y_i' \sim \mathrm{Bin}(20, \pi_i)$ for every posterior draw π_i captures *both* sources, approximating $f(y' \mid y) = \int f(y' \mid \pi) f(\pi \mid y)\, d\pi$.

Answer 8.7

The p-value is not $P(H_0 \mid \textbf{data})$.

- p-value: *assumes* $\pi = 0.2$ and asks how probable data at least this extreme are: $P(\text{data} \mid H_0)$.
- Bayesian: conditions on the observed data and asks how probable the hypothesis is:
  $P(H_0 \mid Y = 14) = 1 - 0.849 = 0.151$.
- These answer different questions and generally differ. The Bayesian answer is the one the student was trying to state.

Answer 8.8

(a) $\text{Beta}(4 + 14,\ 6 + 100 - 14) = \text{Beta}(18,\ 92)$.

(b) Mean $= 18/110 \approx 0.164$; mode $= 17/108 \approx 0.157$.

(c) $E(Y' \mid Y = 14) = 20 \times E(\pi \mid Y = 14) = 20 \times 18/110 \approx 3.27$ (the book's 3.273).

For any π, $E(Y' \mid \pi) = 20\pi$. Averaging over the posterior just replaces π by its posterior mean. That shortcut works for the *mean* only: probabilities and intervals for Y' still need the full posterior predictive model (Q8.4).

Answer 8.9

(a) Prior odds $= 0.2/0.8 = 0.25$; posterior odds $= 0.8/0.2 = 4$; BF $= 4/0.25 = 16$. The data made the hypothesis's odds 16 times larger.

(b) Posterior odds $= 0.094 \times 60 \approx 5.64$.

$$P(H_a \mid Y = 14) = \frac{\text{odds}}{1 + \text{odds}} = \frac{5.64}{6.64} \approx 0.849,$$

matching the book.

Answer 9.1

(a) On average, each 1°F increase in temperature is associated with about 82 more riders per day (80% credible interval: about 76 to 89).

(b) No. $P(\beta_1 > 0) \approx 1$ says the *association* is almost surely positive, not that it is causal. The data are observational: warm days also tend to be in seasons with longer daylight, tourism, holidays and school schedules, any of which could affect ridership. Strong evidence of a relationship is not evidence of cause.

Answer 9.2

(a) No. -2194 is the predicted ridership at 0°F, far outside the observed temperatures in D.C. It is an extrapolation, so β_0 has no practical meaning. The line is only trusted where we have data.

(b) We can reason about β_{0c}: "on a typical (65–70°F) day, ridership is around 5000, likely 3000–7000." Nobody has intuition about rides at 0°F. Centering makes the prior easy to elicit and interpret (and `rstanarm` uses this scale).

Answer 9.3

(a) The credible interval reflects only *posterior variability* (where the line is). The prediction interval adds *sampling variability*: individual days scatter around the line with SD $\sigma \approx 1281$.

(b) The **prediction interval**: the manager faces one specific day, not the long-run average.

(c) The credible interval for the mean shrinks toward a point (the line becomes certain). The prediction interval barely changes: it stays near mean $\pm 1.96\,\sigma \approx \pm 2500$ rides, because day-to-day scatter never goes away. (With 500 days, posterior variability is already small next to σ.)

Answer 9.4

- A slope is learned from *variation in* X. Thirty winter days cover a narrow, cold temperature range, so they say little about what happens on warm days.
- The information in data depends on *what* was observed, not just *how much*.
- Each phase updates the previous posterior. As data span more temperatures, the posterior for β_1 moves and narrows, ending concentrated near 80.
- Updating in phases gives the same final posterior as using all 500 days at once.

Answer 9.5

(a) $100 \pm 2(40) = (20,\ 180)$: each extra degree adds roughly 20 to 180 rides. The prior is sure the relationship is positive but vague about its size.

(b) $1/0.0008 = 1250$ rides: days typically scatter about 1250 rides around the line.

(c) $-2194 + 82.2 \times 75 = -2194 + 6165 = 3971$ rides (the book reports ≈ 3968; the gap is rounding in the coefficients).
A 10°F difference: $82.2 \times 10 = 822$ more rides on the 70°F day.
The intercept cancels.

Answer 9.6

(a)

Draw	$\mu = \beta_0 + 75\beta_1$	$Y' = \mu + \sigma z$
1	$-2200 + 6150 = 3950$	$3950 + 650 = 4600$
2	$-2100 + 6075 = 3975$	$3975 - 1250 = 2725$
3	$-2300 + 6300 = 4000$	$4000 + 256 = 4256$

(b) The μ's build the credible interval for the *average*; the Y''s build the prediction interval. The Y''s are far more spread out (2725 to 4600 vs. 3950 to 4000) because they add day-to-day scatter σz on top of the uncertainty in the line. With 20,000 draws, this is exactly what `posterior_predict()` does.

Answer 15.1

1. **Grouped by school.** Students in one school share teachers, resources and neighborhood.
2. **Grouped by patient.** Repeated measures on one person reflect that person's physiology.
3. **Not grouped.** Independent random sample, one response each.
4. **Grouped by city.** Each city has its own climate (and consecutive days are also correlated over time).

The common thread: knowing one observation from a group tells you something about the others in that group.

Answer 15.2

- Runners differ a lot in baseline speed (between-runner variability). Complete pooling treats those differences as random noise, inflating σ and blurring the age effect.
- Comparisons *across* runners (a fast 60-year-old vs. a slow 55-year-old) are mixed with changes *within* each runner, which is what "slowing with age" actually means.
- Violated assumption: **independence**. Times from the same runner are correlated, and runner identity is ignored.
- Result: a misleading conclusion about the trend and its significance.

Answer 15.3

(a) Five races are very little information. The model extrapolates runner 1's short-term improvement for years, ignoring what the other 35 runners show: people in their 50s–60s tend to slow down. It uses no information from other groups.

(b) It has a separate model for each sampled runner and no model of the *population* of runners. There is nothing to say how a new runner's intercept and slope relate to the others, so it cannot generalize beyond the sample.

Answer 15.4

(a) *Within*: how much one runner's times vary year to year (small for runner 29, large for runner 17). *Between*: how much runners differ from each other in typical speed (29 vs. 17).

(b) Each runner's ability persists: a runner who was fast last year is likely fast this year. So knowing one time tells us about the next.

(c) It models each runner's own trend (respects between-group differences) *and* treats runners as drawn from one population, so each estimate is pulled toward the overall trend. "Learn from one another while celebrating our individuality."

Answer 15.5

(a) Both runners slow down by **+1 min/year**.

(b) $\bar{x} = 58.5$, $\bar{y} = 76$.

- $x - \bar{x}$: $-3.5, -2.5, -1.5, 1.5, 2.5, 3.5$
- $y - \bar{y}$: $4, 5, 6, -6, -5, -4$
- $\sum(x - \bar{x})(y - \bar{y}) = -71, \quad \sum(x - \bar{x})^2 = 41.5$

$b = -71/41.5 \approx -1.71$ min/year: runners appear to get *faster* with age!

(c) The pooled line compares a slow younger runner with a fast older runner: differences *between* runners masquerade as an age effect. Within each runner, time goes up with age. Ignoring the grouping can hide or even reverse a trend (as in Q15.1).

Answer 15.6

(a) $w = 0.5$: $0.5(80) + 0.5(90) = 85$ min.
$w = 0.9$: $0.9(80) + 0.1(90) = 81$ min.

(b) X (2 races) $\rightarrow w = 0.5$; Y (10 races) $\rightarrow w = 0.9$. With more data, a runner's own mean is more reliable, so it gets more weight. With little data, the estimate borrows more from the other runners (it is pulled more toward the overall mean).

(c) $w = 0$: complete pooling (everyone gets 90). $w = 1$: no pooling (each runner keeps their own mean). Partial pooling is everything in between. Chapter 16 shows how w also depends on the within- and between-runner variability.

Multiple-choice answers

MCQ 1: (C) $\hat{R}$. It compares the variability across all chains combined with the variability within each chain. N_{eff}/N, autocorrelation and a single trace plot look at chains one at a time and can miss chains stuck in different places.

MCQ 2: (B) 0.25. $\alpha = \min\{1,\ 0.1/0.4\} = 0.25$. Only the ratio matters; the normalizing constant cancels.

MCQ 3: (C). The prediction interval combines posterior variability and sampling variability. It is *wider* than the credible interval for the mean (A is false), and it never shrinks to a point because days scatter around the line with SD σ (D is false).

MCQ 4: (C). Complete pooling has one universal model; the hierarchical model describes the population of runners. The no-pooled model only knows the 36 sampled runners.